

Architecture Selection & Justification

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We chose Microservices Architecture for the Hostel Maintenance & Issue Ticketing System.

Architectural Choice

The system is built as a set of small, independently deployable services — a Resident Portal and Warden Dashboard as client-facing components, and a Ticket Management Service, Staff Dispatch Service, Notification & Escalation Service, and Ticket Database as backend services, communicating through well-defined provided/required interfaces (see the accompanying component diagram).

Reason 1: Independent Scaling

Ticket creation and photo uploads from the Resident Portal see sharp, unpredictable spikes — for example, when a building-wide water or power outage causes many residents to log tickets within minutes. The Notification & Escalation Service, by contrast, runs a steady, low-volume background check for SLA breaches. Microservices let the Ticket Management Service be scaled up independently to absorb a submission spike without needlessly scaling the escalation or dispatch services, which a single monolithic deployment could not do as cheaply.

Reason 2: Fault Isolation & Independent Deployment

The Notification & Escalation Service depends on third-party SMS/Email gateways that can be slow or temporarily unavailable. Running it as a separate service means a gateway outage degrades only escalation alerts, not the resident's ability to log a new ticket or the warden's ability to update ticket status. Each service can also be updated and redeployed on its own schedule — e.g. adding a new escalation channel — without redeploying or risking downtime for the whole system.

Security Advantage

Because each service exposes only a minimal API surface to the others (e.g. the Resident Portal can reach the Ticket Management Service's CreateTicket interface, but has no direct access to the Ticket Database), a compromise of one client-facing component does not automatically expose the underlying ticket data store. Access to resident-submitted photos and room numbers can be scoped and encrypted entirely within the Ticket Management Service and Ticket Database, limiting the blast radius of any single breach.

Performance Benefit

The Notification & Escalation Service polls the Ticket Database and checks SLA timers asynchronously, in the background, decoupled from the request path of ticket creation or status updates. This keeps the resident-facing “log a ticket” flow fast and responsive even while thousands of open tickets are simultaneously being monitored for 24-hour SLA breaches, since that monitoring work never blocks the main service's request/response cycle.